

COMPUTER NETWORKS — WEEK 3 CHEAT SHEET

LAN Interconnection · Hubs · Bridges · Spanning Tree · Switches · FDDI | Instructor: D. MSI

1. Why Not One Big LAN?

- **Limited traffic capacity:** more devices share same bandwidth, speed drops
- **Limited cable length:** Ethernet has a max cable distance
- **Large collision domain:** more devices = more collisions
- **Limited stations:** Token Ring gets slow with too many stations

Collision domain = a segment where two devices sending at once can collide. Smaller domains = fewer collisions. Bridges/switches shrink collision domains.

Devices by OSI Layer

Device	Layer	Forwards by
Repeater/Hub	1 (Physical)	No intelligence, just signal
Bridge/Switch	2 (Data Link)	Destination MAC address
Router	3 (Network)	Destination IP address

2. Repeaters and Hubs

Repeater: Layer 1. Just amplifies electrical signal — no understanding of frames. Classic Ethernet allows 4 repeaters, stretching cable from 500m to 2500m.

Hub: multi-port repeater. Sends signal to all ports (broadcasts). Can be arranged in tiers (multi-tier hub) with a backbone hub on top.

Hub Pros

- Cheap and simple
- Graceful degradation — one hub failing doesn't kill everything
- Extends distance ~100m per hub

Key limit: Hubs do NOT split collision domains — all ports share ONE big domain. That's why switches replaced them.

3. Bridges

Layer 2 device. Reads destination MAC address and forwards smartly — only to the right segment, not everywhere.

- **Selective forwarding:** only sends to the segment where destination lives
- **Collision isolation:** each port = its own collision domain (big plus over hubs)
- **Buffering:** stores frames temporarily to control traffic
- Uses CSMA/CD when sending onto a segment
- Built with 4-8 line cards (one per input/output line)

Backbone bridge: one central bridge joins multiple LANs. Without it: single point of failure + traffic bottleneck through one segment.

4. Bridge Learning & Filtering

Bridges use **self-learning** to build a table: MAC address → port.

Example: C sends frame to D

- Bridge has no entry for D → floods frame to all ports
- Bridge learns: C arrived on Port 1 → records C→Port 1
- D replies → bridge learns D→Port 2
- Next time, frame for C goes only to Port 1 (no flooding)

Problem: Loops

Two bridges connecting same two LANs = loop. Causes wrong learning, duplicate frames, and **broadcast storms** (frames loop forever).

Fix: Build a loop-free tree so there's only one path between any two nodes.

5. Spanning Tree Protocol (STP)

Redundant paths are good for reliability but cause loops. STP disables some interfaces to remove loops while keeping the network connected.

- **Root Bridge:** the elected center of the tree
- **BPDU:** Bridge Protocol Data Unit — messages bridges use to elect root and pick which ports to disable
- **Designated Port:** active port on a segment
- **Blocked Port:** disabled to stop loops
- **Standard:** IEEE 802.1D

Remember: STP allows exactly ONE active path between any two LANs.

6. Bridges vs. Routers

Both are **store-and-forward** — they fully receive data before sending it on.

Feature	Bridge	Router
Layer	2	3
Reads	MAC address	IP address
Table	Filtering table	Routing table
Loop fix	Spanning tree	TTL + routing protocols
Broadcast storm	Forwarded (no protection)	Blocked (firewall-like)
Setup	Plug and play	Needs IP config
Best for	Small networks (~200 hosts)	Large networks (1000s)

7. Ethernet Switches

Advanced Layer 2 device — like a bridge, but connects individual computers, not whole segments.

- Forwards by MAC address, just like a bridge
- Many devices can talk at the same time, no collisions
- Every port = its own collision domain
- Star topology — each host has its own cable to the switch

Cut-Through Switching

Switch starts forwarding as soon as it reads the destination MAC — doesn't wait for the whole frame. Lowers delay, good for real-time traffic.

Dedicated vs Shared Ports

Dedicated: full bandwidth for one device only. **Shared:** bandwidth split among a small group (still better than a hub).

Switch vs Bridge

Feature	Bridge	Switch
Connects	LAN segments	Individual computers
Ports	Few (4-8)	Many (12-48+)
Forwarding	May skip same segment	Must forward all frames
Buffer need	Lower	Higher

8. FDDI — Fiber Distributed Data Interface

Long-distance LAN standard using optical fiber. Copper version = CDDI.

- **Speed:** 100 Mbit/s (up to 200 with both rings)
- **Range:** up to 200 km (single ring), 100 km (dual ring)
- **Frame size:** up to 4352 bytes (vs Ethernet's 1500)
- Based on IEEE 802.4 (Token Bus) — NOT 802.5 (Token Ring)

Station Types

- **DAS** (Dual-Attached): connects to both rings, for critical devices
- **SAS** (Single-Attached): connects to one ring via a concentrator

Dual ring: primary ring carries traffic; secondary ring is backup, takes over if primary fails. Typical design = "dual ring of trees".

FDDI Frame Fields

PA(preamble) → SD(start) → FC(frame control) → DA(dest MAC) → SA(source MAC) → PDU(data) → FCS(checksum) → ED/FS(end/status)

9. Ethernet vs Token Ring — Access Methods

Condition	Ethernet (CSMA/CD)	Token Ring
Low load	Excellent, fast	Good
High load	Degrades (collisions)	Excellent (no collisions)
Latency	Unpredictable	Predictable
QoS support	None	Yes, can prioritize

Ethernet: devices compete to send; collision = both stop, wait random, retry. Token: only token holder sends — no collisions, but more complex.

Rule of thumb: Ethernet for low-medium load. Token passing for guaranteed, predictable access.

Quick Exam Recall

Device	Layer	Collision Domain
Hub	1	One shared for all ports
Bridge	2	One per port
Switch	2	One per port
Router	3	One per port

- Hubs = no collision isolation. Bridges/switches DO isolate.
- STP disables ports, doesn't remove physical cables
- FDDI secondary ring is idle unless primary fails
- Cut-through switching = lower latency, forwards before full frame arrives
- Token passing has zero collisions; Ethernet handles collisions via CSMA/CD

বল স্কলন: একটি বড় LAN ব্যবহারিক নয় কারণ ব্যান্ডউইথ ভাগ, তার দৈর্ঘ্য ও সংখ্যের সীমা থাকে। Hub সংকেত পাঠায় সবার কাছে, কোনো বুদ্ধিমত্তা নেই। Bridge ও Switch MAC ঠিকানা দেখে বুদ্ধিমত্তার মতো পাঠায় এবং প্রতিটি পোর্ট আলাদা সংখ্যক ভোয়েইন। Spanning Tree গুলি বন্ধ করে। FDDI দ্রুতগতির ফাইবার নেটওয়ার্ক, দুটি রিং ব্যবহার করে নিরাপত্তার জন্য।